NCERT Summary: Topic 20: Environment & Sustainable Development — Raw Notes

Source Integration

- Class 11 Indian Economic Development, Chapter 7: Environment and Sustainable Development

Core Factual & Conceptual Architecture

1. Environment and Carrying Capacity

- Definition: Totality of biotic (living) and abiotic (non-living) elements forming our planetary inheritance.
- Four Vital Functions: Supplies resources (renewable and non-renewable), assimilates waste, sustains life diversity, and provides aesthetic services.
- Carrying & Absorptive Limits: The environmental crisis occurs when human resource extraction exceeds regeneration rates and waste generation overwhelms absorptive capacity, leading to high economic and health opportunity costs.

2. State of India's Environment

- Demographic Intensity: Supporting 17 percent of humanity and 20 percent of livestock on 2.5 percent of global land places extreme stress on finite natural capital.
- Environmental Dichotomy: Concurrent existence of poverty-induced environmental degradation (survival pressure) and pollution driven by affluence and industrial growth.
- Five Priority Concerns: Land degradation (soil erosion of 5.3 billion tonnes annually), biodiversity loss, urban vehicular air pollution, fresh water management, and solid waste management.
- Institutional Response: Central Pollution Control Board (CPCB), established in 1974, sets standards and monitors air and water quality nationwide, supported by grassroots movements like Chipko and Appiko.

3. Sustainable Development Concepts and Criteria

- Brundtland Definition: Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.
- Barbier's Standard: Emphasizes raising the material standard of living of the poor (improving income, health, education) while minimizing environmental degradation.
- Herman Daly's Conditions: Limit human population within carrying capacity, drive input-efficient technology, keep renewable extraction within regeneration rates, limit non-renewable depletion to renewable substitution rates, and correct pollution inefficiencies.

4. Strategies for Sustainable Development

- Non-Conventional Energy: Transitioning away from polluting thermal and ecologically disruptive hydro power toward rural LPG and gobar gas (biogas), urban CNG, wind energy, and solar photovoltaic systems.
- Traditional Wisdom: Reintegrating indigenous practices that treat humanity as an intrinsic part of the environment rather than its master.

✍ Notes & Updates